

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

May 2012

EE 203 Electrical Measurement and Measuring Instruments

Date: 17.05.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Write a short note on Alternating Field type moving iron power factor meter. [04]
(b) Explain briefly electrical resonance type frequency meter. [03]
- Q - 2 (a) Draw the phasor diagram of Potential Transformer (P.T.). Also derive the equation for transformation ratio and phase angle for P.T. [07]
(b) Explain in detail location of earth fault and short-circuit fault for underground cables by Varley loop test. [07]

OR

- Q - 2 (a) A current transformer of turns ratio 1:199 is rated as 1000/5 A, 25 VA. The core-loss and magnetizing component of primary current are 4 and 7 A under rated conditions. Determine the phase angle and ratio errors for the rated burden and rated secondary current at 0.8 p.f. lagging and 0.8 p.f. leading. Neglect the resistance and leakage reactance of secondary winding. [07]
(b) Explain in detail location of earth fault and short-circuit fault for underground cables by Murray loop test. [07]
- Q - 3 (a) Derive the general torque equation of Moving Iron type instrument. [04]
(b) Explain construction and operating principle of Electrodynamometer type instrument [06]
(c) Explain the operation of full-wave rectifier type voltmeter. [04]

OR

- Q - 3 (a) State and explain advantages and disadvantages of Permanent Magnet Moving coil type instruments [04]
(b) Explain in detail with a neat diagram working of Repulsion type Moving Iron instrument. [06]
(c) Derive the force and torque equations of electrostatic instruments. [04]

SECTION - II

- Q - 4 (a) What is limiting error? Derive the expression for limiting error. [04]
(b) Define accuracy and precision related to measuring instruments. Also explain difference between them. [03]
- Q - 5 (a) Discuss in detail working of Crompton's potentiometer. [04]
(b) Explain the calibration of ammeter with the help of D.C. potentiometer. [03]
(c) Explain briefly Direct deflection method and Loss of charge method for measurement of high order resistance. [07]

OR

- Q - 5 (a) State different types of A.C. Potentiometer. Explain in detail any one of them. [07]
(b) Explain briefly substitution method for measurement of medium order resistance. [04]
(c) Write a Short note on Megaohm bridge. [03]
- Q - 6 (a) Explain with a neat sketch and phasor diagram Hay's bridge. Also state its advantages and disadvantages. [05]
(b) The four arms of a bridge as shown in Fig. 1. are: [04]

Arm ab: an imperfect capacitor C_1 with an equivalent series resistance of r_1 .

Arm bc: a non-inductive resistance R_3 ,

Arm cd: a non-inductive resistance R_4

Arm da: an imperfect capacitor C_2 with an equivalent series resistance of r_2 in series with a resistance R_2

A supply of 450 Hz is given between terminal a and c and the detector is connected between b and d. At balance:

$R_2 = 4.8 \Omega$, $R_3 = 2000 \Omega$, $R_4 = 2850 \Omega$ and $C_2 = 0.5 \mu\text{F}$ and $r_2 = 0.4 \Omega$. Calculate the value of C_1 and r_1 and also of the dissipating factor for this capacitor.

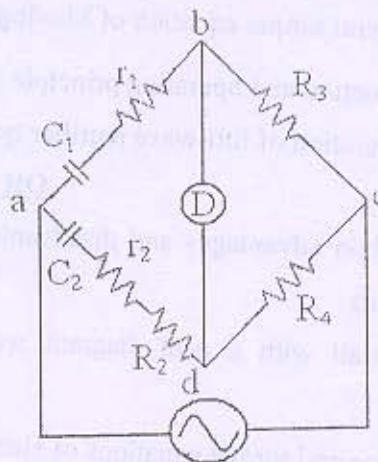


Fig. 1. Bridge

- (c) Explain with a neat diagram step by step method for determination of B-H curve for a specimen. [05]

OR

- Q - 6 (a) Explain with a neat sketch and phasor diagram Schering bridge. [05]
- (b) A Maxwell's Inductance-Capacitance bridge shown in Fig.2 is used to measure an unknown inductance in comparison with capacitance. The various values at balance: [04]

$$R_2 = 400\Omega; R_3 = 600\Omega; R_4 = 1000\Omega; C_4 = 0.5\mu F.$$

Calculate the values of R_1 and L_1 . Calculate also the value of storage (Q) factor of coil if frequency is 1000 Hz.

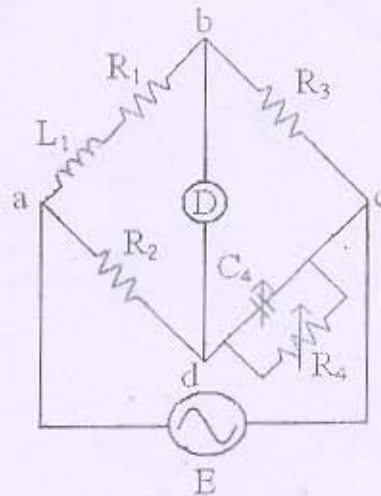


Fig. 2. Maxwell's Inductance-Capacitance Bridge

- (c) Write a short note on method of reversals for determination of hysteresis loop. [05]
